

L9a: Introduction to Derivatives and European Option Pricing

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Objectives for Today

We introduce derivatives and begin our discussion of derivatives pricing. What is a payment that may never arrive worth today?

Today's concepts

- Define and classify derivatives: futures, forwards, swaps, and options, with their underlying assets and market structures.
- Analyze option payoffs and profits: payoff and profit for long and short calls and puts, and how the strike and premium affect profitability at expiration.
- Price European options contracts: the premium makes the expected net present value zero under the risk-neutral pricing rule. We compute it by Monte Carlo simulation and with the Black–Scholes–Merton formulas, and relate calls and puts with put–call parity.

Lectures and Examples

Lecture:
CHEME-5660-L9a-Lecture-IntroductionToDerivatives-BSM-Fall-2026.ipynb

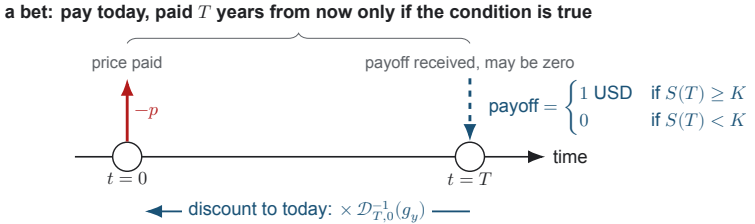
Example 1:
CHEME-5660-L9a-Example-SingleContractPayoffProfit-Fall-2026.ipynb

Example 2:
CHEME-5660-L9a-Example-BSM-Premium-Fall-2026.ipynb

The first example computes payoff and profit for single long and short call and put contracts. The second computes European call and put premiums by Monte Carlo simulation and with the Black–Scholes–Merton formulas, then looks at how the premium changes with strike price.

Concept Review: A Bet on the Future

Suppose we pay p today for a contract that pays 1 USD in T years if $S(T) \geq K$, and nothing otherwise.



Discounting the payoff at the benchmark growth rate g_y gives a random NPV:

$$\text{NPV}_0 = -p + \mathcal{D}_{T,0}^{-1}(g_y)\, V(S(T)) = \begin{cases} -p + e^{-g_y T} & \text{if } S(T) \geq K \text{ (true)} \\ -p & \text{if } S(T) < K \text{ (false)} \end{cases}$$

Expected NPV of the Bet

Under our beliefs about $S(T)$, the expected NPV is:

$$\mathbb{E}[\text{NPV}_0] = (-p + e^{-g_y T}) \underbrace{\mathbb{P}(S(T) \geq K)}_{\text{true}} + (-p) \underbrace{\mathbb{P}(S(T) < K)}_{\text{false}} \qquad \text{weight by probability}$$

$$= -p \underbrace{[\mathbb{P}(S(T) \geq K) + \mathbb{P}(S(T) < K)]}_{=1} + e^{-g_y T} \mathbb{P}(S(T) \geq K) \qquad \text{collect the } p \text{ terms}$$

$$= -p + e^{-g_y T} \underbrace{\mathbb{P}(S(T) \geq K)}_{\text{true}} \qquad \text{probabilities sum to one}$$

The bet has a non-negative expected NPV when $p \leq e^{-g_y T} \mathbb{P}(S(T) \geq K)$.

The bet is a derivative contract: its payoff depends on the future price of another asset (or event). Every buyer pays one market price, yet buyers with different beliefs value the same contract differently.

How is the market price set?

Company Profile: Cboe Global Markets

Cboe operates markets for options, equities, futures, and foreign exchange. It opened on April 26, 1973, as the Chicago Board Options Exchange, the world's first exchange for listed options.

- Listed options: standard strikes, expirations, and contract sizes, so the same contract can trade many times a day. The Options Clearing Corporation (OCC) stands between the buyer and the seller.
- Index options and volatility: SPX options are European-style and cash-settled. The VIX reads the market's expected 30-day volatility of the S&P 500 from their premiums.
- Event contracts: since June 2026, binary options on the Mini-S&P 500 that pay 100 USD or nothing, cleared by OCC.

Today's question: where does a listed premium come from?

What Is a Derivative?

A derivative is a contract between a buyer and a seller to transact an asset at a set price on a future date. It is an abstract asset whose cash flows depend on an underlying asset or event.

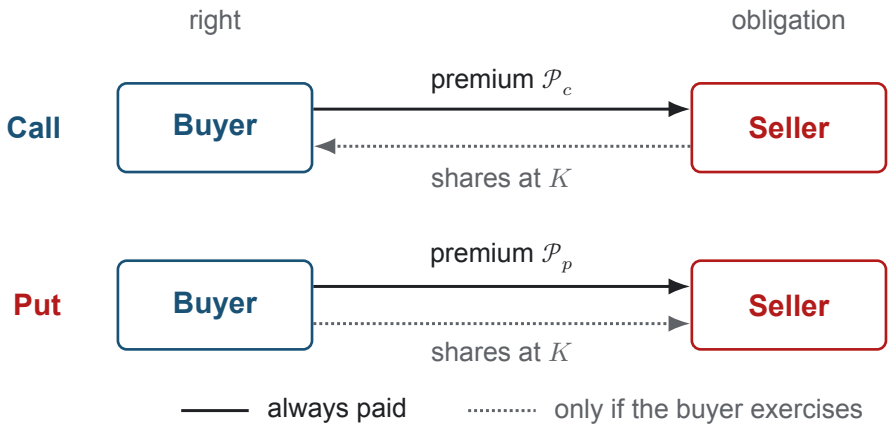
- Futures, forwards, and swaps: obligate both parties to transact. Futures trade on exchanges with standard terms. Forwards trade over the counter, where each party bears the other's default risk. Swaps exchange one asset's cash flows for another's, also over the counter.
- Options: the buyer holds the right, but not the obligation, to buy (call) or sell (put) at a set price. Options are exchange-traded.
- Event contracts: pay a fixed amount, usually one dollar, if an event occurs and nothing otherwise. Kalshi lists them.

How Big Is the US Options Market?

- 2025: a record of about 15.2 billion contracts, or about 60.6 million per day (OCC).
- 2026: through September, about 13.2 billion contracts, or about 70.6 million per day (OCC).
- Notional: about 4 trillion USD per day in 2025, more than three times the 1.1 trillion USD per day traded in U.S. cash equities (Traders Magazine, Cboe).

Sources: the lecture notebook's market table, with dates and links.

Call and Put Options Contracts



The Business Case for Long and Short Calls

A call option gives the owner the right, but not the obligation, to buy 100 shares of an underlying asset (per contract) at the strike price K by the expiration date.

- Buyer (long): benefits from upside price movement without purchasing the asset directly. Long calls have defined risk: the maximum loss is the premium paid. If the owner exercises, they buy shares at K instead of the market price.
- Seller (short): receives the premium. Short calls have unlimited upside risk, so calls are often sold only by investors who already own the shares, as a covered call (100 shares per contract). Short calls obligate the seller to sell shares at K if the option is exercised.

The Business Case for Long and Short Puts

A put option gives the owner the right, but not the obligation, to sell 100 shares at the strike price K by the expiration date.

- Buyer (long): benefits from downside price movement without selling the asset directly. Long puts have defined risk: the maximum loss is the premium paid. The owner can sell shares at K instead of the market price.
- Seller (short): receives the premium. Short puts have large but bounded downside risk: if the share price falls to zero, the seller loses the strike price minus the premium's future value per share. Puts are often sold only by investors with the cash to buy the shares (100 per contract), a cash-secured put. Short puts obligate the seller to buy shares at K if the option is exercised.

Call and Put Payoffs

The payoff per share of a single call or put contract at expiration (or when the contract is exercised) is:

$$\boxed{\begin{aligned} V_c(K, S(T)) &= \max(S(T) - K,\ 0) = (S(T) - K)^+ \\ V_p(K, S(T)) &= \max(K - S(T),\ 0) = (K - S(T))^+ \end{aligned}}$$

where K is the strike price and $S(T)$ is the share price at expiration (time $t = T$). The seller charges the buyer a premium per share, $\mathcal{P}_c$ for a call and $\mathcal{P}_p$ for a put.

Profit at Expiration

The premium is paid at $t = 0$ and the payoff arrives at $t = T$, so we compare them at expiration using the premium's future value, $\mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}$, where $\mathcal{D}_{T,0}(g_y) = e^{g_y T}$. The buyer's profit per share for a single contract is:

$$P_c^{\text{buyer}}(K, S(T)) = V_c(K, S(T)) - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c$$
$$P_p^{\text{buyer}}(K, S(T)) = V_p(K, S(T)) - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p$$

A position's orientation θ is $+1$ for the buyer (long) and -1 for the seller (short). Its profit per share is $\theta \cdot P^{\text{buyer}}$, so the seller's profit is $P^{\text{seller}} = (-1) \cdot P^{\text{buyer}}$, and $P^{\text{buyer}} + P^{\text{seller}} = 0$ at every share price.

Options are a zero-sum game: one party's gain is the other's loss.

Breakeven

Setting the buyer's profit to zero gives the breakeven share price $S^{\ddagger}$ at expiration:

$$S^{\ddagger} = K + \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c \qquad \text{call}$$

$$S^{\ddagger} = K - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p \qquad \text{put}$$

Above the call breakeven, and below the put breakeven, the buyer profits and the seller loses. Listed contracts often expire within weeks, so $\mathcal{D}_{T,0}(g_y)$ is close to one (about 1.003 for 30 days at $g_y = 4\%$).

Example 1: CHEME-5660-L9a-Example-SingleContractPayoffProfit-Fall-2026.ipynb

Pricing Put and Call Options Contracts

An option's price depends on the underlying asset's value, strike price, time to expiration, and exercise style.

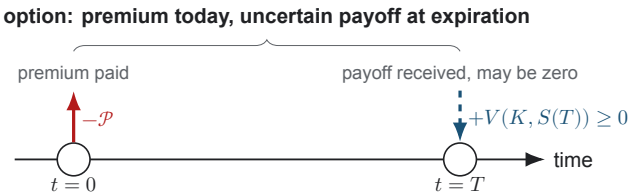
- European-style: exercise only at expiration.
- American-style: exercise any time through expiration.
- Bermudan-style: less common, exercise on specified dates.

The American option's added flexibility makes it worth at least as much as an otherwise identical European option.

Options on single stocks and on exchange-traded funds such as SPY are American-style. The European-style contracts we can trade are cash-settled index options: SPX (SPXW for weekly expirations), XSP, NDX, and RUT.

Let's start with European-style options and compute their net present value.

Options as Abstract Assets



The buyer pays the premium $\mathcal{P}$ today and receives the payoff $V(K, S(T)) \geq 0$ at expiration, so the net present value at $t = 0$ is:

$$\text{NPV}_0 = -\mathcal{P} + \mathcal{D}_{T,0}^{-1}(g_y) \cdot V(K, S(T))$$

Factoring out $\mathcal{D}_{T,0}^{-1}(g_y)$ gives $\text{NPV}_0 = \mathcal{D}_{T,0}^{-1}(g_y) \cdot P^{\text{buyer}}$: the NPV is the buyer's profit at expiration discounted to today, so it is zero at the breakeven share price.

The European Premium

How is the market price set? A call is a bet whose payoff grows with the share price. It pays $S(T) - K$ instead of 1 USD when $S(T) > K$, and nothing otherwise. Following the same steps as for the bet:

$$\begin{aligned}
\mathbb{E}[\mathrm{NPV}_0] &= \left(-\mathcal{P} + \mathcal{D}_{T,0}^{-1}(g_y)\,\bar{V}\right)\mathbb{P}(S(T) > K) + (-\mathcal{P})\,\mathbb{P}(S(T) \le K) && \text{weighted sum} \\
&= -\mathcal{P}\,\underbrace{\left[\mathbb{P}(S(T) > K) + \mathbb{P}(S(T) \le K)\right]}_{=1} + \mathcal{D}_{T,0}^{-1}(g_y)\,\bar{V}\,\mathbb{P}(S(T) > K) && \text{collect } \mathcal{P} \text{ terms} \\
&= -\mathcal{P} + \mathcal{D}_{T,0}^{-1}(g_y)\,\mathbb{P}(S(T) > K)\,\bar{V} && \text{sum to one}
\end{aligned}$$

where $\bar{V}$ is the average payoff $S(T) - K$ over the outcomes with $S(T) > K$. The expected NPV is zero when the premium is the discounted chance of exercise times the average payoff when exercised.

Whose Probabilities?

$\mathbb{P}$ holds our own beliefs about the future share price $S(T)$, so this premium is the most we would pay. The market prices every option with one shared risk-neutral pricing rule $\mathbb{Q}$, so there is one price. Under $\mathbb{Q}$, the expected share price grows at the risk-free rate g_y, and GBM with no dividends has drift g_y. Repeating the calculation under $\mathbb{Q}$:

$$\begin{aligned}
\mathbb{E}_{\mathbb{Q}}[\mathrm{NPV}_0] &= \left(-\mathcal{P} + \mathcal{D}_{T,0}^{-1}(g_y)\,\bar{V}\right)\mathbb{Q}(S(T) > K) + (-\mathcal{P})\,\mathbb{Q}(S(T) \le K) \\
&= -\mathcal{P}\ \underbrace{\left[\mathbb{Q}(S(T) > K) + \mathbb{Q}(S(T) \le K)\right]}_{=1} \\
&\qquad + \mathcal{D}_{T,0}^{-1}(g_y)\ \underbrace{\left[\bar{V}\,\mathbb{Q}(S(T) > K) + 0 \cdot \mathbb{Q}(S(T) \le K)\right]}_{=\mathbb{E}_{\mathbb{Q}}[V_c(K, S(T))]} \\
&= -\mathcal{P} + \mathcal{D}_{T,0}^{-1}(g_y)\,\mathbb{E}_{\mathbb{Q}}\left[V_c(K, S(T))\right]
\end{aligned}$$

where $\bar{V}$ is now the average payoff over the outcomes with $S(T) > K$ under $\mathbb{Q}$.

The Market Premium

The second bracket is the expected payoff of the call under $\mathbb{Q}$: the payoff in each outcome, weighted by its probability. Setting $\mathbb{E}_{\mathbb{Q}}[\mathrm{NPV}_0] = 0$ gives the market premium of the call, $\mathcal{P}_c = \mathcal{D}_{T,0}^{-1}(g_y)\,\mathbb{E}_{\mathbb{Q}}[V_c(K, S(T))]$.

A put, which pays $K - S(T)$ when $S(T) < K$, gives the same result with V_p in place of V_c. For either contract, the premium is the discounted expected payoff under $\mathbb{Q}$:

$$\boxed{\mathcal{P}^{\mathsf{EU}} = \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{E}_{\mathbb{Q}}[V(K, S(T))]}$$

Event Contracts

The Concept Review bet is an event contract on $S(T) \geq K$, which pays 1 USD if the event occurs and nothing otherwise. Its market price is:

$$p = \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{E}_{\mathbb{Q}}[\mathbb{1}\{\text{event}\}] = \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{Q}(\text{event})$$

The pricing probability $\mathbb{Q}(\text{event})$ can differ from the real-world probability $\mathbb{P}(\text{event})$. A general Kalshi event, such as an election, is not tied to a share price, so choosing its pricing probability requires additional assumptions.

A Monte Carlo Premium

Simulation gives a direct way to estimate the market premium. The payoff depends only on $S(T)$, so we draw it from the solution of GBM with drift g_y.

Initialize: given $S(0)$, K, the payoff function $V(K, S)$ of the call or put, T (years), σ, g_y, and the number of simulated paths M.

1. For each path $i = 1, 2, \ldots, M$ **do**:
   a. Draw a standard normal random number Z_i.
   b. Compute the share price at expiration: $S_i(T) \leftarrow S(0) \cdot \exp[(g_y - \sigma^2/2)\,T + \sigma\sqrt{T}\,Z_i]$.
   c. Compute the payoff: $V_i \leftarrow V(K, S_i(T))$.
2. The premium is the discounted average payoff: $\mathcal{P} \leftarrow \mathcal{D}_{T,0}^{-1}(g_y) \cdot \frac{1}{M} \sum_{i=1}^{M} V_i$.

Optional: an alternative way to think about the premium, in the contingent claims notebook.

The Black–Scholes–Merton Equation

Merton and Scholes received the 1997 Nobel Prize in Economic Sciences for this model. Fischer Black had passed away before the prize was awarded.

Assume the price of a non-dividend-paying share follows GBM with constant volatility σ, and the risk-free rate g_y is constant. Write the premium at time t, when the share price is S, as $\mathcal{P}(S,t)$, with $\mathcal{P}(S,T) = V(K,S)$. It satisfies:

$$\boxed{\frac{\partial \mathcal{P}}{\partial t} + g_y S \frac{\partial \mathcal{P}}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 \mathcal{P}}{\partial S^2} = g_y \mathcal{P}}$$

Only g_y and σ appear, not the real-world growth rate of the share price. Solving backward from the payoff gives $\mathcal{P}(S(0),0) = \mathcal{P}^{\text{EU}}$, the premium the Monte Carlo simulation estimates.

Derivation: the contingent claims notebook.

Black–Scholes–Merton Premiums

Instead of simulating, we can compute this premium with the closed-form European call and put formulas:

$$\boxed{\begin{aligned}\mathcal{P}_c &= N(d_+)\cdot S(0) - N(d_-)\cdot K\cdot \mathcal{D}_{T,0}^{-1}(g_y)\\ \mathcal{P}_p &= N(-d_-)\cdot K\cdot \mathcal{D}_{T,0}^{-1}(g_y) - N(-d_+)\cdot S(0)\end{aligned}}$$

where N is the standard normal CDF, T is in years, and:

$$d_+ = \frac{1}{\sigma\sqrt{T}}\left[\ln\frac{S(0)}{K} + \left(g_y + \frac{\sigma^2}{2}\right)T\right], \qquad d_- = d_+ - \sigma\sqrt{T}$$

Historical or Implied Volatility?

There are two ways to put a number on σ, and the second also gives a quick range for the share price:

- Historical volatility: previously, we estimated σ from past share prices as $\sigma_{\text{ann}} = \sqrt{\Delta t} \cdot \sigma_g$, where σ_g is the standard deviation of the growth rates and Δt is the time between price observations. This estimate looks backward.
- Implied volatility: the implied volatility (IV) is the market's forecast of a likely movement in the share price, the volatility "implied" by an option's market price through a pricing model such as Black–Scholes–Merton. It is often equal to or larger than the historical volatility.
- How can we use it? IV represents a one-standard-deviation movement in the share price over the next year. Over a horizon of T years, the size of that move scales with $\sqrt{T}$.

A Back-of-the-Envelope Range from IV

Moving one standard deviation up or down in the solution of geometric Brownian motion gives a back-of-the-envelope range for the share price:

$$
\begin{aligned}
S(T) &= S(0)\exp\left[\mu_g T + \sigma\sqrt{T}\,Z\right] && \text{solution of geometric Brownian motion} \\
S_{\pm} &= S(0)\exp\left[\mu_g T \pm \mathsf{IV}\sqrt{T}\right] && \text{set } \sigma = \mathsf{IV} \text{ and } Z = \pm 1 \\
&\approx S(0)\exp\left[\pm \mathsf{IV}\sqrt{T}\right] && \mu_g T \text{ is small next to } \mathsf{IV}\sqrt{T} \text{ for short } T \\
&\approx S(0)\left(1 \pm \mathsf{IV}\sqrt{T}\right) && e^x \approx 1 + x \text{ for small } x
\end{aligned}
$$

Here μ_g is the mean growth rate, Z is a standard normal random variable, and $S_{\pm}$ is the share price one standard deviation above or below $S(0)$. For example, a share at $S(0)$ = 100 USD/share with IV = 40% and 30 days to expiration (T = 30/365 years) has $\mathsf{IV}\sqrt{T}$ = 0.115, so the range is about 88.5 to 111.5 USD/share. Under the model, the share price ends inside this range about 68% of the time.

Which σ?

If we knew the right volatility, the formula would return the market premium. In practice, we run it backward. Like a bond's yield to maturity, the implied volatility $\hat{\sigma}$ is the σ at which the BSM premium matches the market premium $\mathcal{P}^{\text{market}}$.

Initialize: given $\mathcal{P}^{\text{market}}$, $S(0)$, K, T (years), g_y, a bracket $\sigma_{\text{lo}} < \sigma_{\text{hi}}$ with $\mathcal{P}(\sigma_{\text{lo}}) \le \mathcal{P}^{\text{market}} \le \mathcal{P}(\sigma_{\text{hi}})$, a tolerance $\epsilon > 0$, `maxiter`, and $i \leftarrow 1$.

While $\sigma_{\text{hi}} - \sigma_{\text{lo}} > \epsilon$ and $i \le$ `maxiter` **do**:

1. Compute $\sigma_{\text{mid}} \leftarrow (\sigma_{\text{lo}} + \sigma_{\text{hi}})/2$ and its BSM premium $\mathcal{P}(\sigma_{\text{mid}})$.
2. If $\mathcal{P}(\sigma_{\text{mid}}) < \mathcal{P}^{\text{market}}$, **then** $\sigma_{\text{lo}} \leftarrow \sigma_{\text{mid}}$. Otherwise, $\sigma_{\text{hi}} \leftarrow \sigma_{\text{mid}}$.
3. Update the iteration counter $i \leftarrow i + 1$.

The implied volatility is the midpoint of the final bracket.

Implied Volatility in Practice

- One answer: the BSM premium rises with σ, so exactly one σ matches the market premium, and each step halves the bracket around it.
- Why the formula: the Monte Carlo premium changes with the random draws and costs M simulated paths at every step.
- One per contract: each strike and expiration has its own implied volatility, although the BSM model assumes one constant σ. To price a contract without a quote, or on a future date, we can predict it from a surface fitted to today's quotes (Sun et al., 2026).

Optional: the SPXW volatility skew notebook applies this bisection to a chain of S&P 500 index options.

Put–Call Parity for European Contracts

For a European call and put with the same strike and expiration, the payoffs differ by $S(T) - K$ at every share price. Subtracting the put premium from the call premium gives:

$$
\begin{aligned}
\mathcal{P}_c - \mathcal{P}_p &= \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{E}_{\mathbb{Q}}[V_c(K, S(T)) - V_p(K, S(T))] && \text{premium formula} \\
&= \mathcal{D}_{T,0}^{-1}(g_y) \cdot \mathbb{E}_{\mathbb{Q}}[S(T) - K] && \text{payoff identity} \\
&= \mathcal{D}_{T,0}^{-1}(g_y) \cdot (S(0)\, e^{g_y T} - K) && \text{expected share price grows at } g_y \\
&= S(0) - K \cdot \mathcal{D}_{T,0}^{-1}(g_y) && \mathcal{D}_{T,0}^{-1}(g_y) = e^{-g_y T}
\end{aligned}
$$

This is put–call parity: $\boxed{\mathcal{P}_c - \mathcal{P}_p = S(0) - K \cdot \mathcal{D}_{T,0}^{-1}(g_y)}$

Parity holds only for European contracts. An American holder can exercise before expiration, so the difference between the call and put premiums is bounded, not fixed.

Example 2: CHEME-5660-L9a-Example-BSM-Premium-Fall-2026.ipynb

What About American-Style Options?

The holder can exercise at any time $\tau \leq T$, chosen using only the information available at that time. The premium is the largest expected payoff, discounted from τ back to today, over all such exercise times:

$$\mathcal{P}^{\mathsf{AM}} = \max_{\tau \leq T} \mathbb{E}_{\mathbb{Q}}\left[\mathcal{D}_{\tau,0}^{-1}(g_y) \cdot V(K, S(\tau))\right] \geq \mathcal{P}^{\mathsf{EU}}$$

The American put has no closed-form formula like BSM, so its premium is usually computed on a binomial lattice.

Optional Advanced Material

Contingent claims: advanced/contingent_claims/
CHEME-5660-L9a-Advanced-ContingentClaims-BSM-Fall-2026.ipynb

We derive where the BSM premiums come from: the risk-neutral probability on a one-step lattice, state prices for event contracts, the BSM equation and its solution, and the call built from a digital claim and an asset-or-nothing claim. We also compare a client's certainty equivalent for the bet with its market price.

Volatility skew: advanced/spxw_volatility_skew/
CHEME-5660-L9a-Advanced-SPXW-Volatility-Skew-Fall-2026.ipynb

We apply the implied-volatility bisection to a synthetic chain of SPXW options and compare the implied volatility with the realized volatility of the index.

Notebooks: lectures/week-9/L9a/advanced, also linked from the lecture notebook.

Summary

In this lecture, we explored derivatives, a fundamental asset class used for risk management, hedging, and speculation.

Key takeaways

- Derivatives derive their value from underlying assets: common types are futures, forwards, swaps, and options, each with different characteristics, maturities, and uses.
- Option payoffs are determined by strike prices and contract orientation: payoff diagrams at expiration show profitability and risk exposure.
- The premium makes the expected NPV zero under the risk-neutral pricing rule: Monte Carlo simulation or the BSM formulas compute it for European contracts, and put–call parity links calls and puts.